TrES-5b

R-1738

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-5_130628` · created 2026-08-02T18:17:31+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456471.9348 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.14 +/- 0.05
Transit depth (Rp/Rs)^2	1.97 +/- 1.41 [%]
Orbital Inclination (inc)	83.43 +/- 2.76
Airmass coefficient 1 (a1)	0.713 +/- 0.034
Airmass coefficient 2 (a2)	0.29 +/- 0.31
Scatter in the residuals of the lightcurve fit is	4.77 %
Best Comparison Star	#2 - [153, 116]
Optimal Aperture	3.36
Optimal Annulus	15.42
Transit Duration (day)	0.061 +/- 0.023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.14 ± 0.05 vs 0.142 (deviation 0.04σ)
Duration fit vs published	0.061 d vs 0.0758 d (deviation 0.64σ)
Mid-time O–C	3.4 min
Depth SNR	2.8
Residual scatter	4.77 %

Light curve

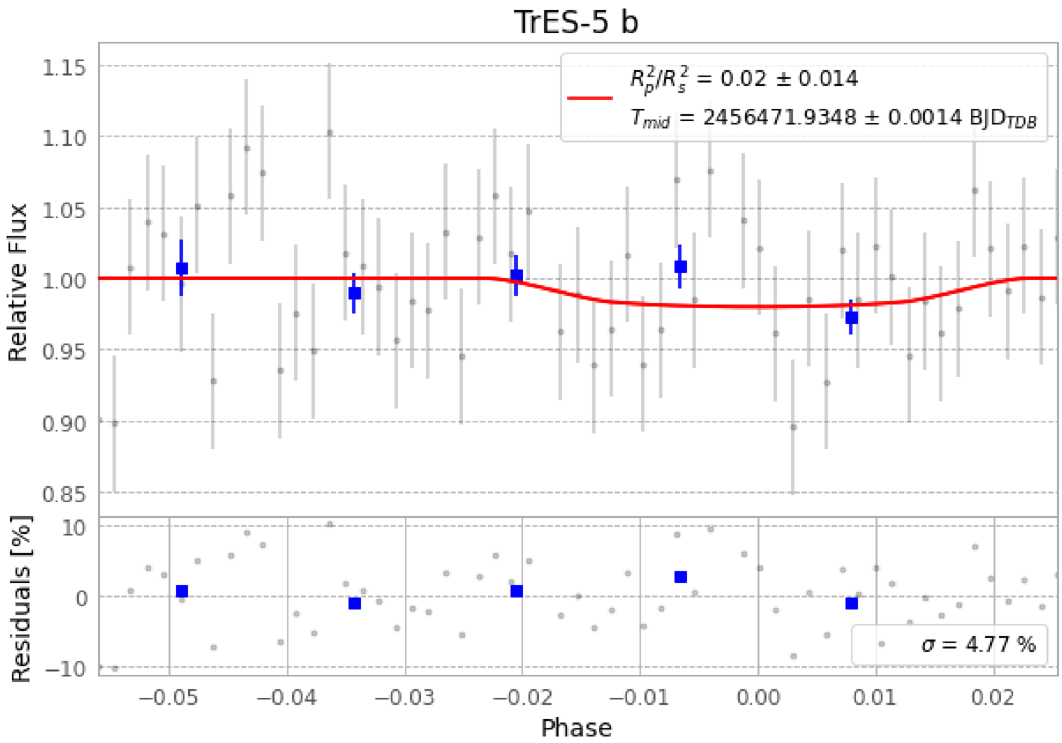

Figure 1 — FinalLightCurve_TrES-5 b_2013-06-28.png (SHA-256 4b488853f63960f2…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [446, 260], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 87689a74d36d8f66…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

59 FITS frames (39 MB), `TRES-5130628082413.FITS` → `TRES-5130628111812.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-130628.log (c74fab29a759…), FinalLightCurve_TrES-5 b_2013-06-28.png (4b488853f639…), FinalLightCurve_TrES-5 b_2013-06-28.csv (e624146e7bc8…)

Compute resources

Wall time 160.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-08-02 18:17Z.